ALMA MATER STUDIORUM
UNIVERSITÀ DI BOLOGNA

ETH zürich

Combining Global and Local Perception for Autonomous Navigation on Nano-UAVs

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European Robotics Forum

EUROPEAN
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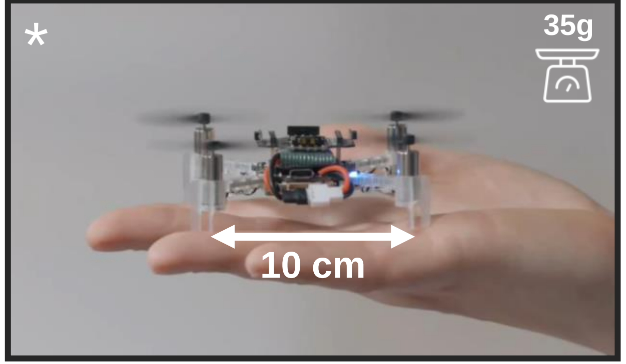
Autonomous nano-drones: applications

Nano-UAV



* Image from McGuire et al., “Minimal navigation solution for a swarm of tiny flying robots to explore an unknown environment.” Sci. Robot.4 (2019).
+ image from Technology Innovation Institute (TII), Abu Dhabi

Autonomous nano-drones: applications

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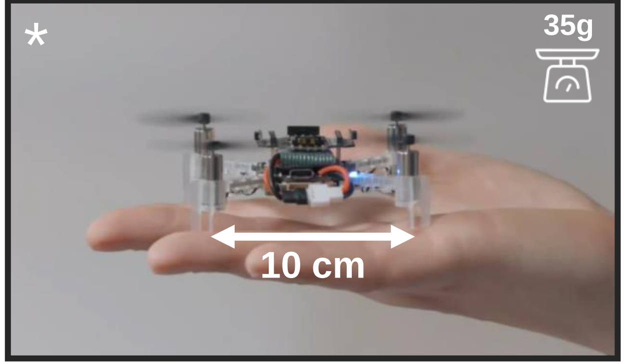
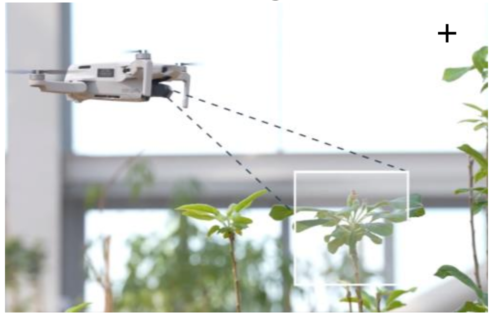


Applications

Surveillance

Precision agriculture



Entertainment

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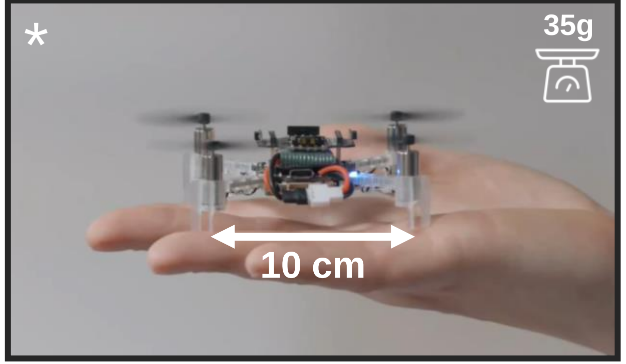

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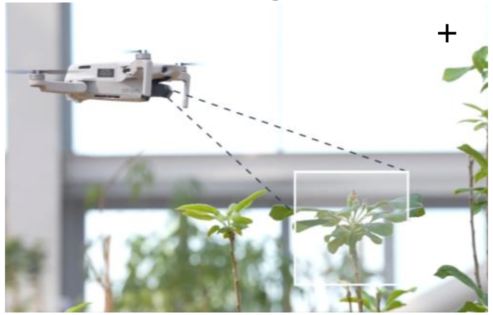

Entertainment

Required skills

Autonomous navigation

Perception	Sensors	Visual cues	Precise geometry

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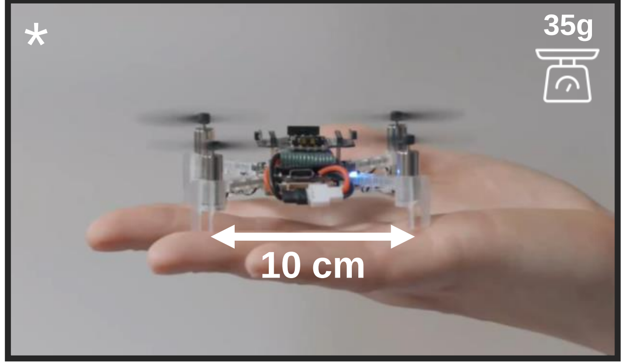

Applications

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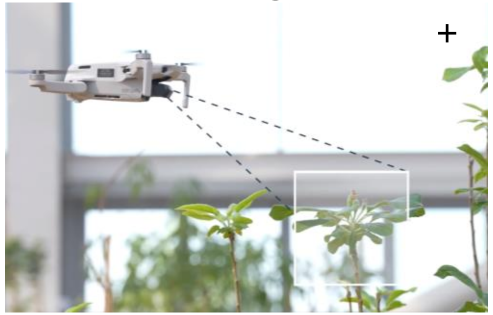

Entertainment

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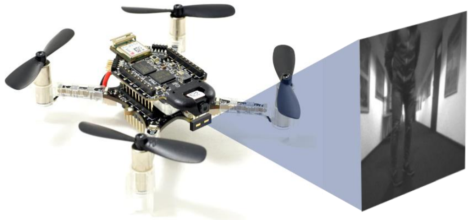

Perception	Sensors	Visual cues	Precise geometry
Global	Vision-based	✓	✗

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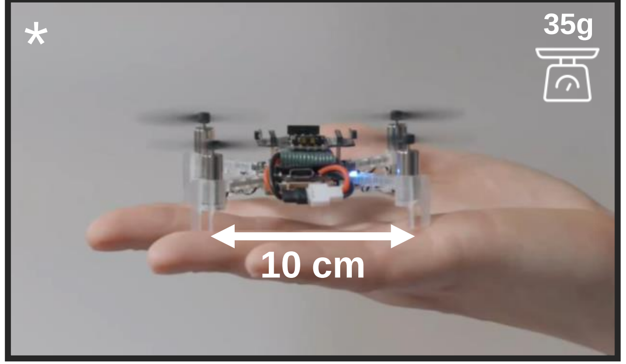
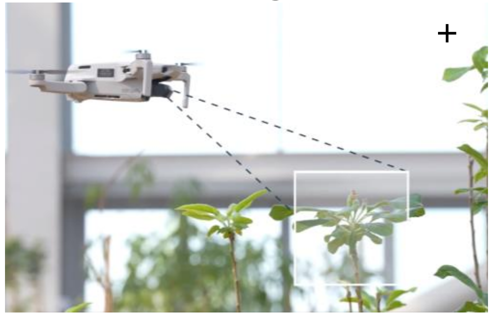


Applications

Surveillance

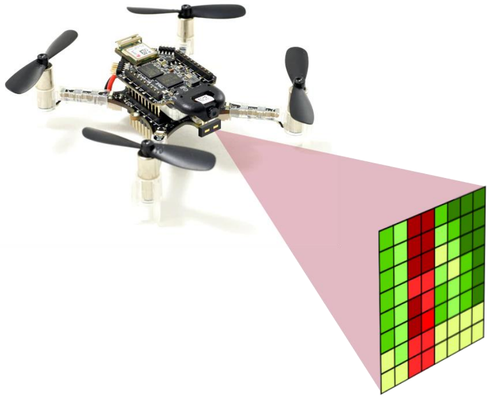
Precision agriculture



Entertainment

Required skills

Autonomous navigation



Perception	Sensors	Visual cues	Precise geometry
Global	Vision-based	✓	✗
Local	Depth-based	✗	✓

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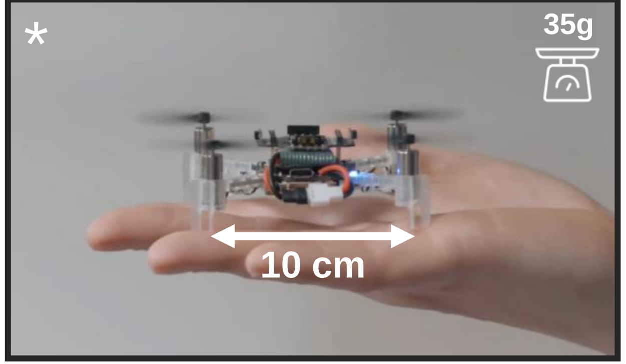

Applications

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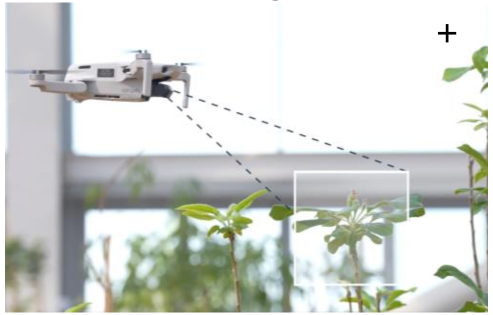

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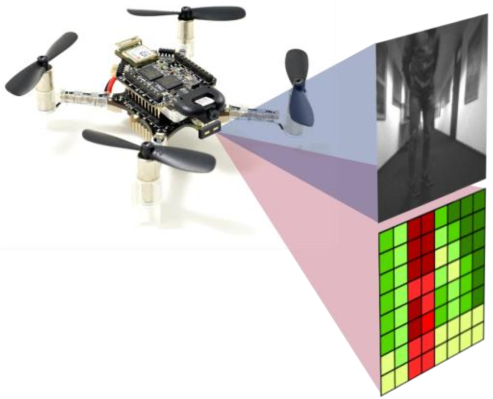

Perception	Sensors	Visual cues	Precise geometry
Global	Vision-based	✓	✗
Local	Depth-based	✗	✓
Global+Local	Vision-depth	✓	✓

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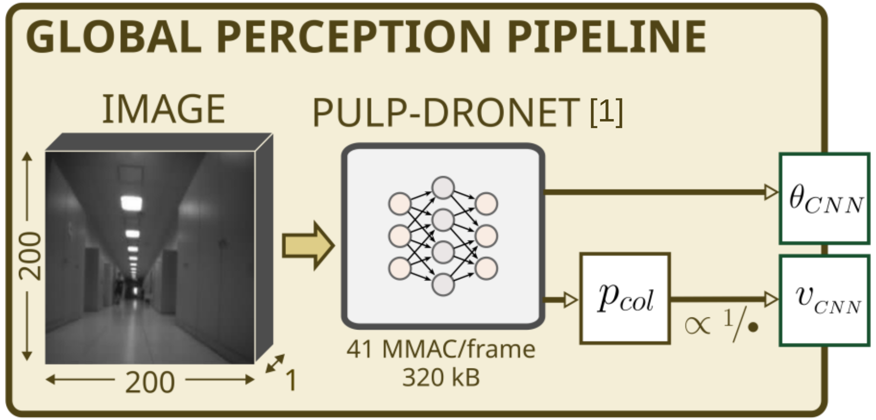
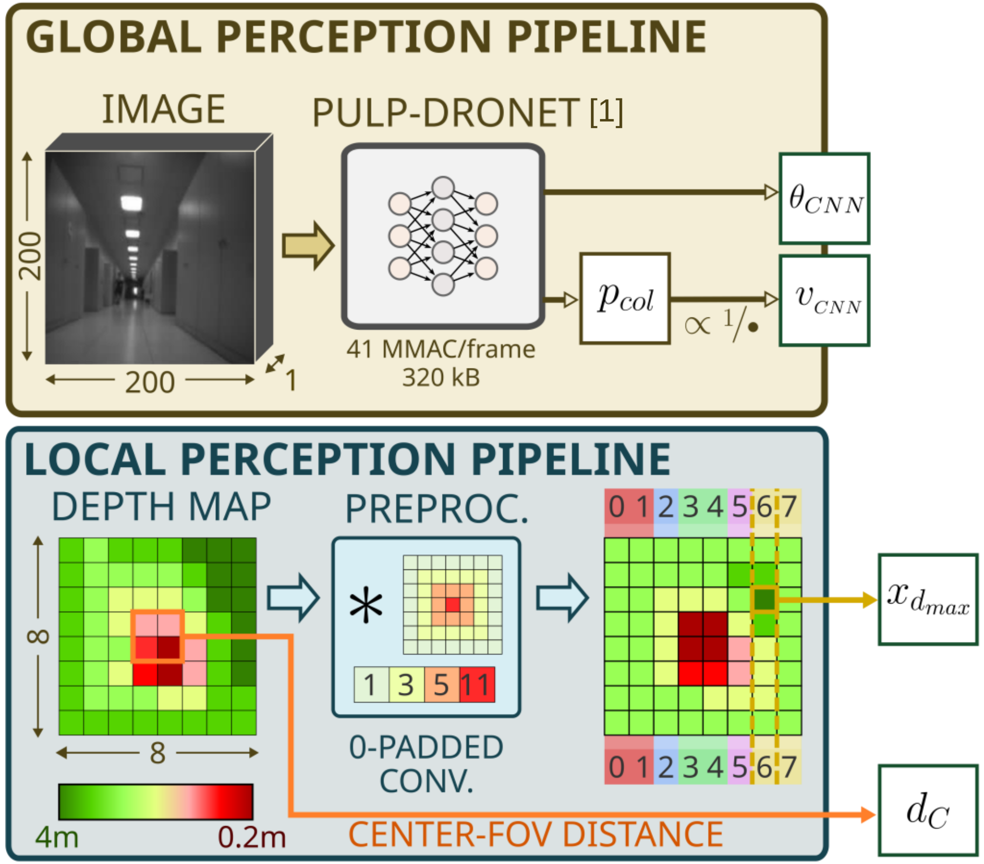
Global + Local perception pipeline



[1] Palossi *et al.*, "A 64-mW DNN-Based Visual Navigation Engine for Autonomous Nano-Drones," in IEEE Internet of Things Journal, vol. 6, no. 5, pp. 8357-8371, Oct. 2019,
[2] H. Müller *et al.*, "Robust and Efficient Depth-Based Obstacle Avoidance for Autonomous Miniaturized UAVs," in *IEEE Transactions on Robotics*, vol. 39, no. 6, pp. 4935-4951, Dec. 2023

Global + Local perception pipeline



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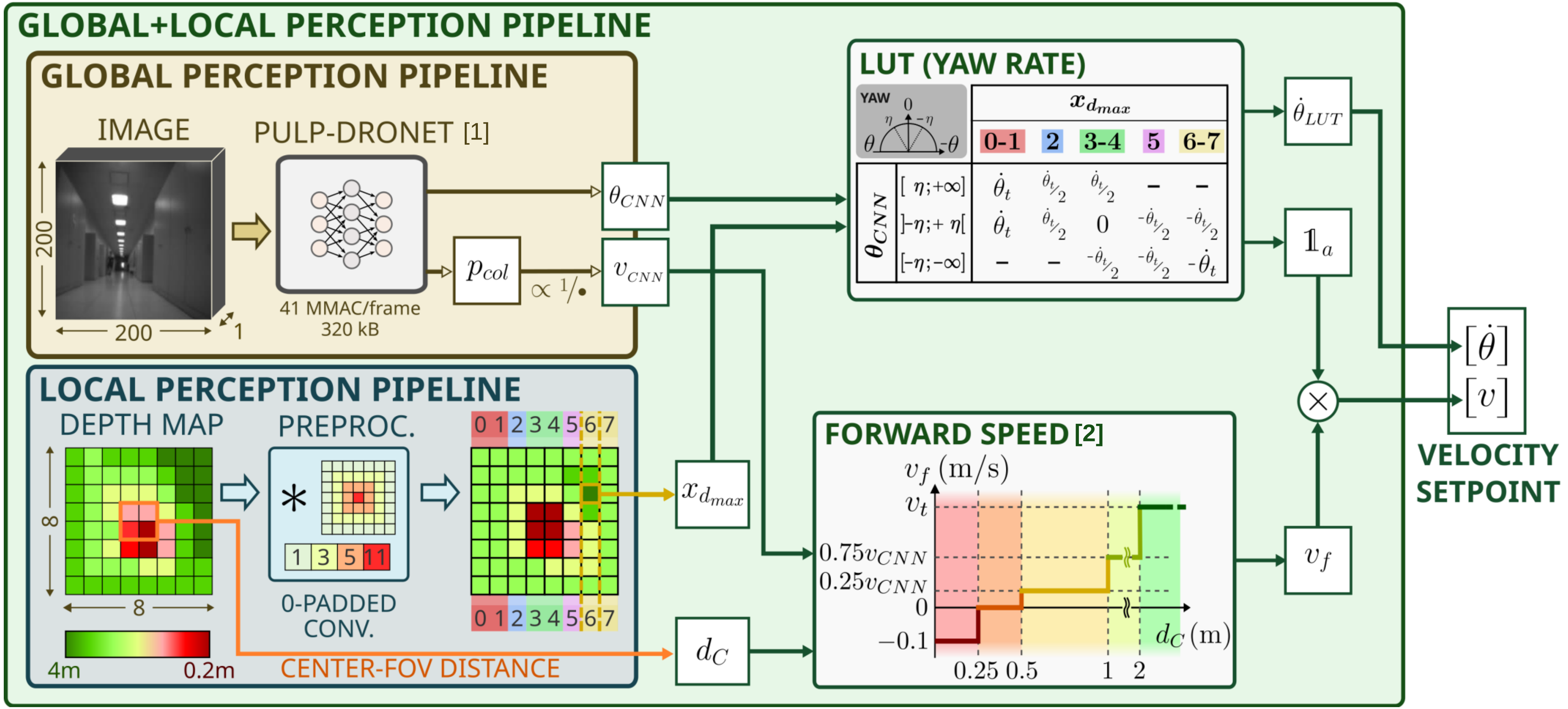

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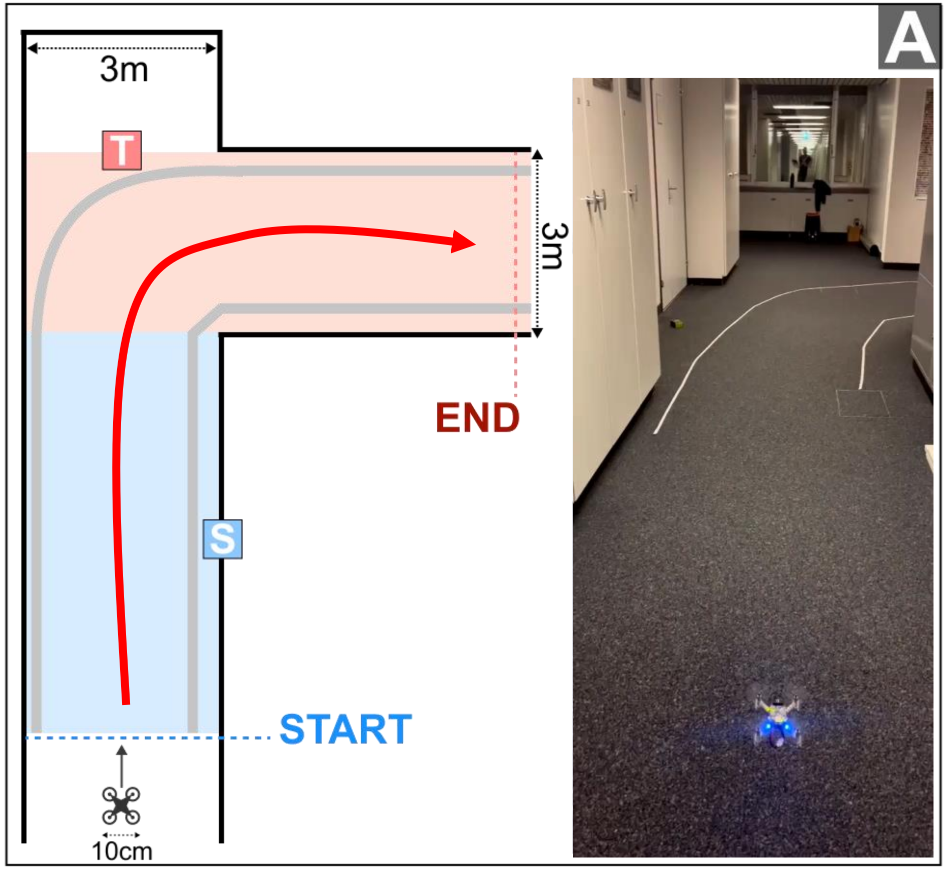
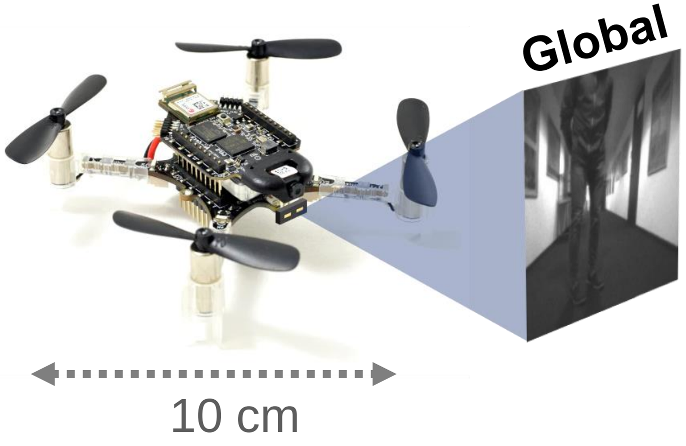
Results

Scenario A: no obstacles



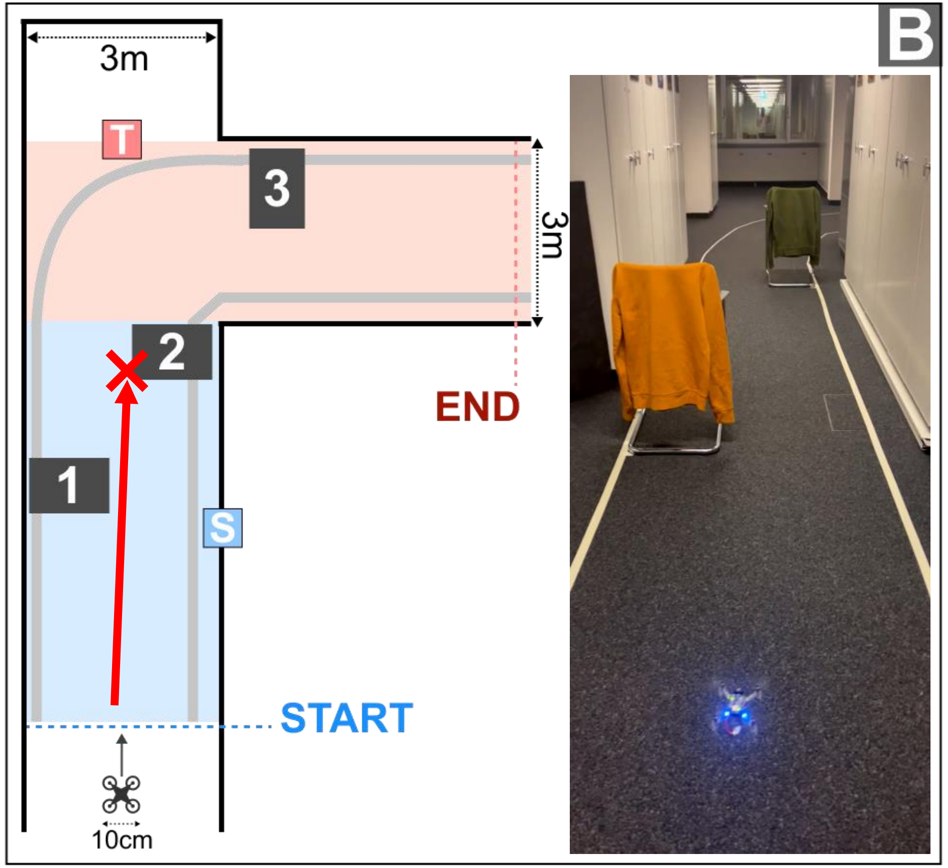
Straight segment · Turn segment · Obstacle · Floor lines



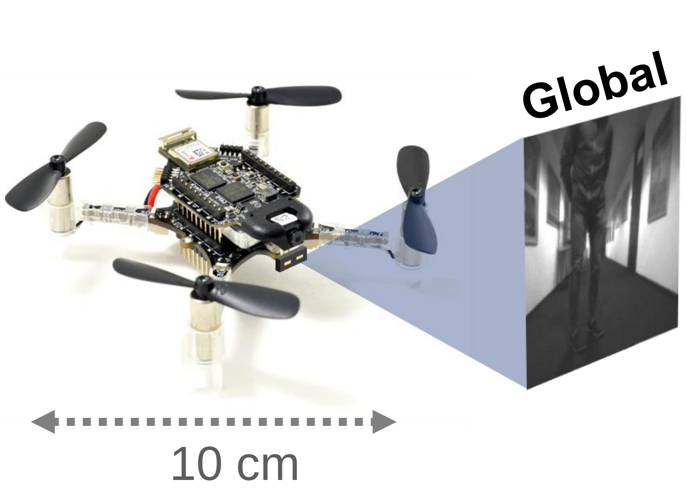
Global perception	
Turns (visual cues)	Static obstacles
✓	

Results

Scenario B: static obstacles + turn



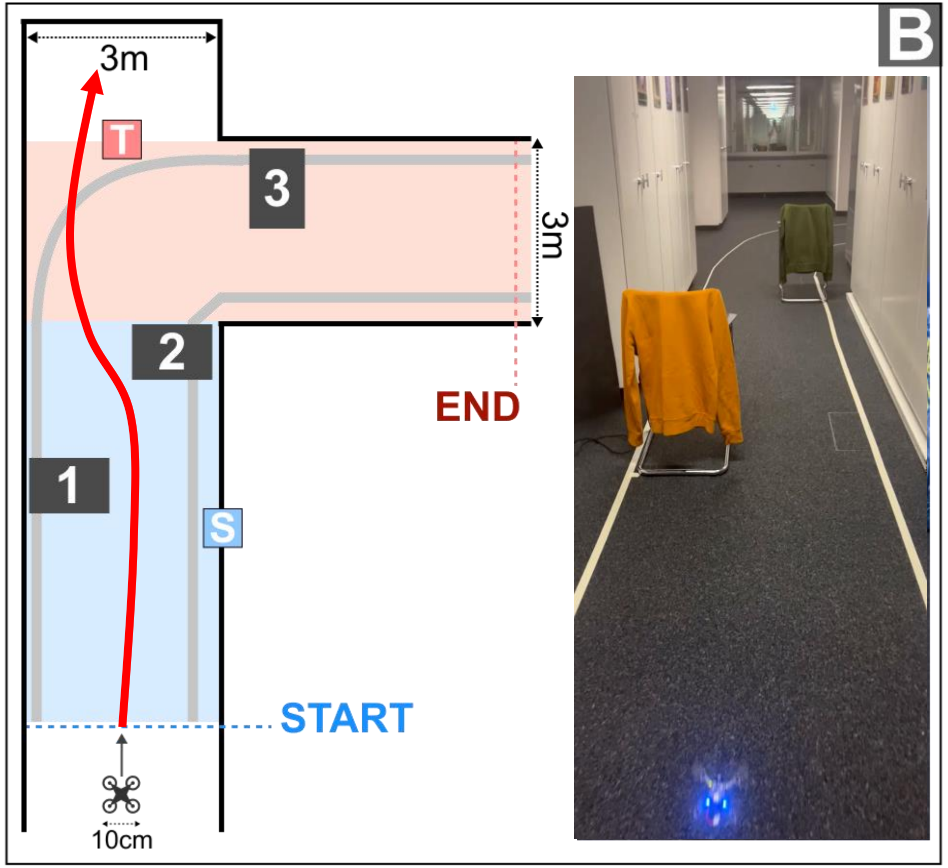
S Straight segment T Turn segment Obstacle Floor lines

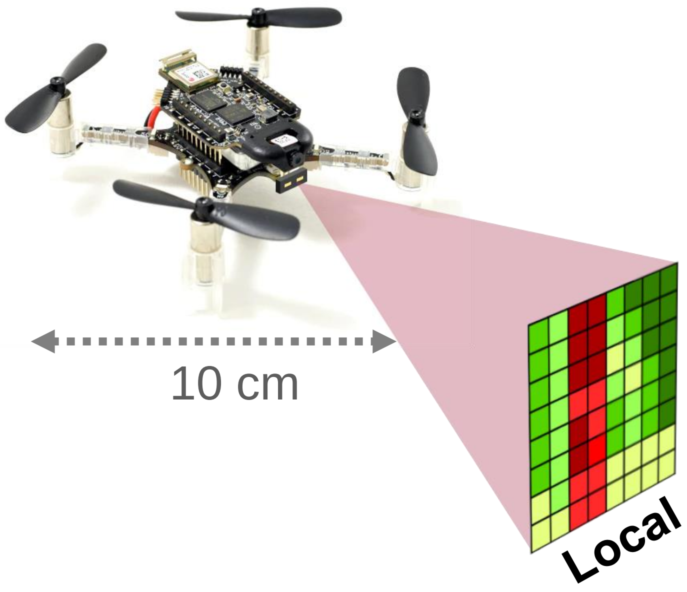


Global perception	
Turns (visual cues) ✓	Static obstacles ✗

Results

Scenario B: static obstacles + turn





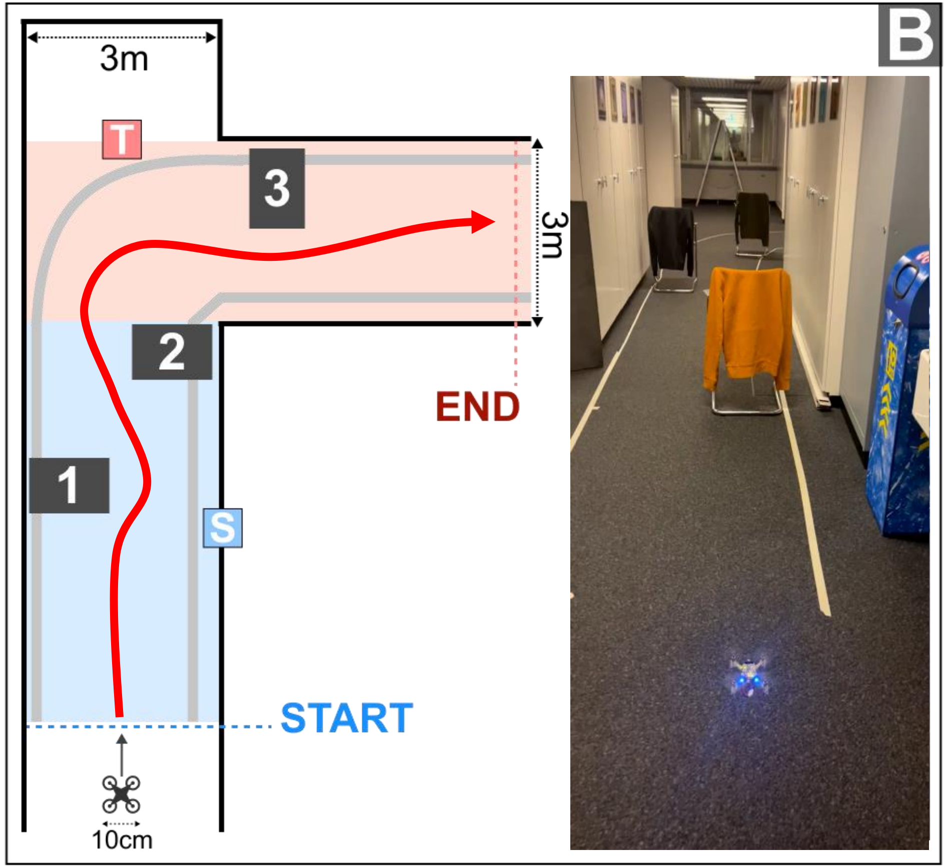
Global perception	
Turns (visual cues)	Static obstacles
✓	✗

Local perception	
Turns (visual cues)	Static obstacles
✗	✓

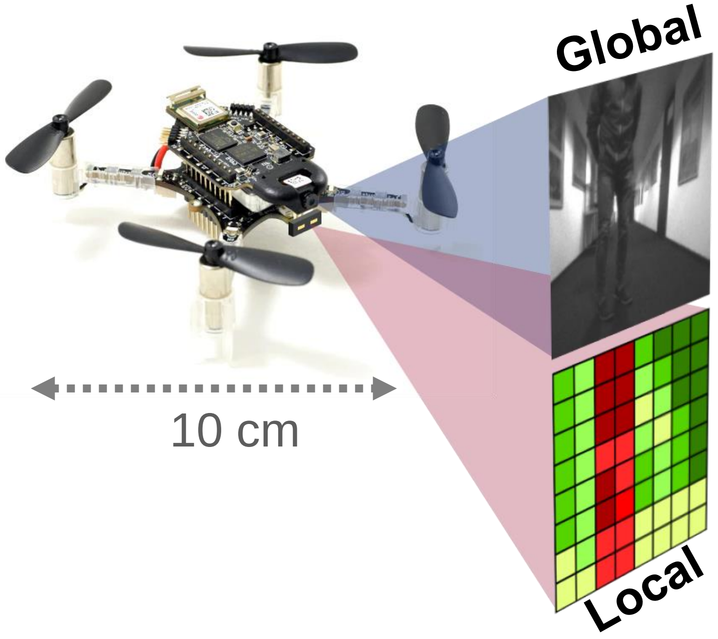
S Straight segment T Turn segment ■ Obstacle ▬ Floor lines

Results

Scenario B: static obstacles + turn



S Straight segment   T Turn segment   ■ Obstacle   ▬ Floor lines



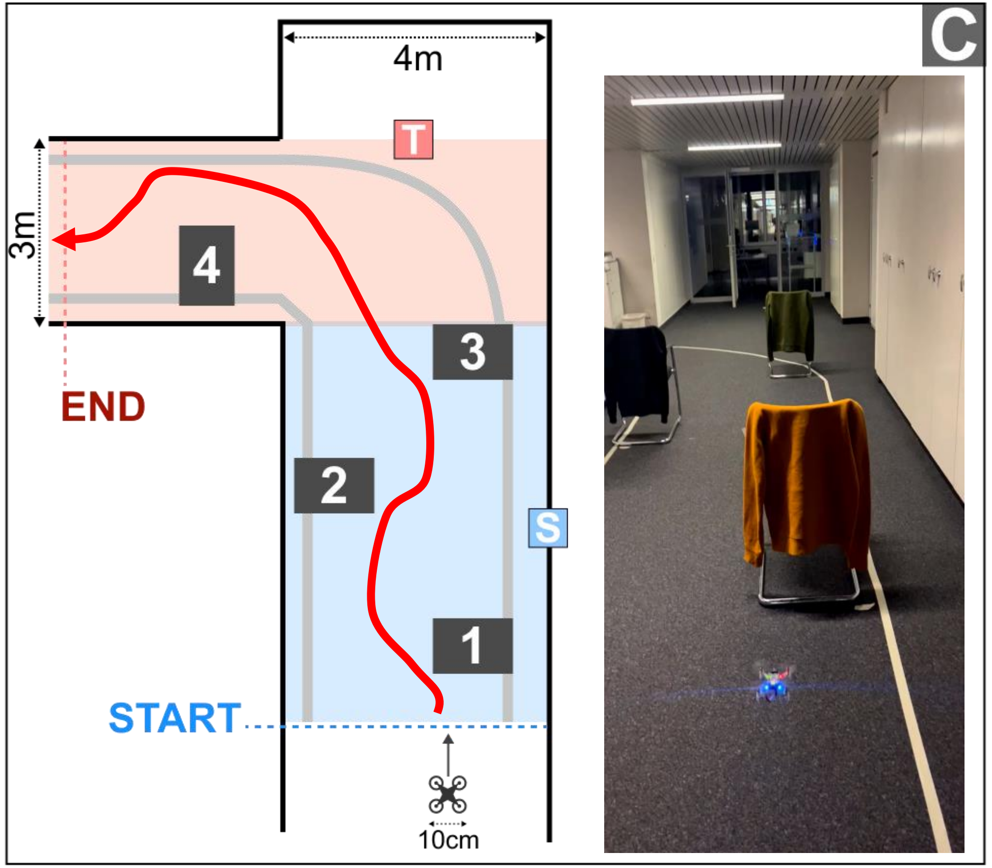
Global perception	
Turns (visual cues)	Static obstacles
✓	✗

Local perception	
Turns (visual cues)	Static obstacles
✗	✓

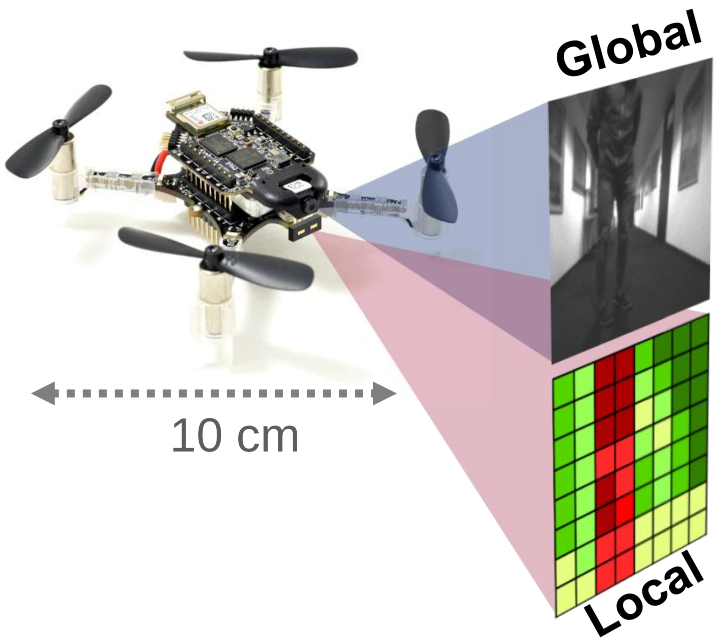
Global + local perception	
Turns (visual cues)	Static obstacles
✓	✓

Results

Scenario C: static obstacles + turn



S Straight segment T Turn segment Obstacle Floor lines



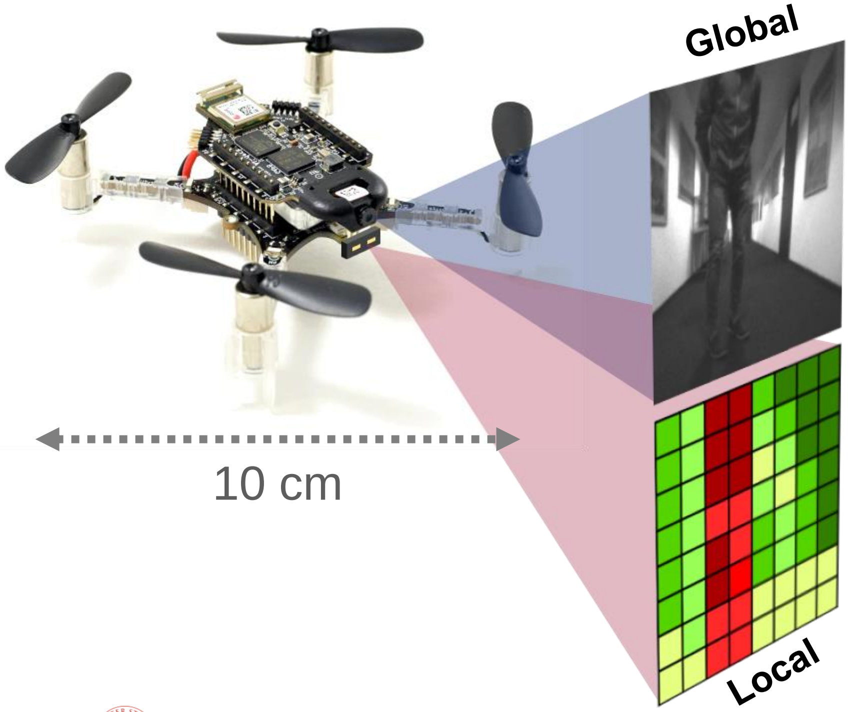
Global perception	
Turns (visual cues)	Static obstacles
✓	✗

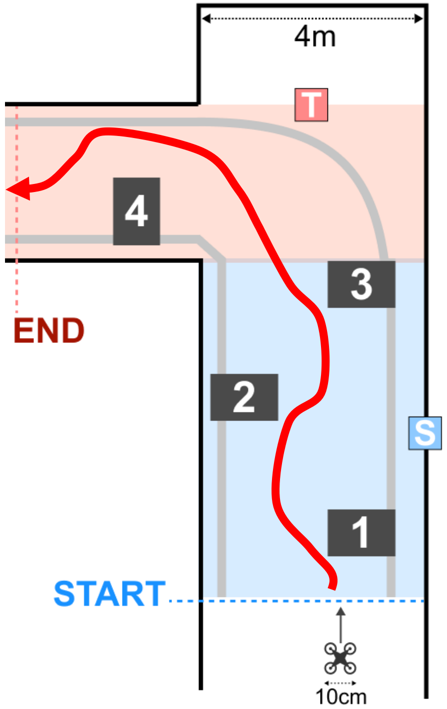
Local perception	
Turns (visual cues)	Static obstacles
✗	✓

Global + local perception	
Turns (visual cues)	Static obstacles
✓	✓

Conclusion

Platform	Task	Methodology	In-field results
Nano-UAV	Autonomous navigation	Global perception: vision-based; Local perception: depth-based.	Turns (visual features only); Static obstacle avoidance.





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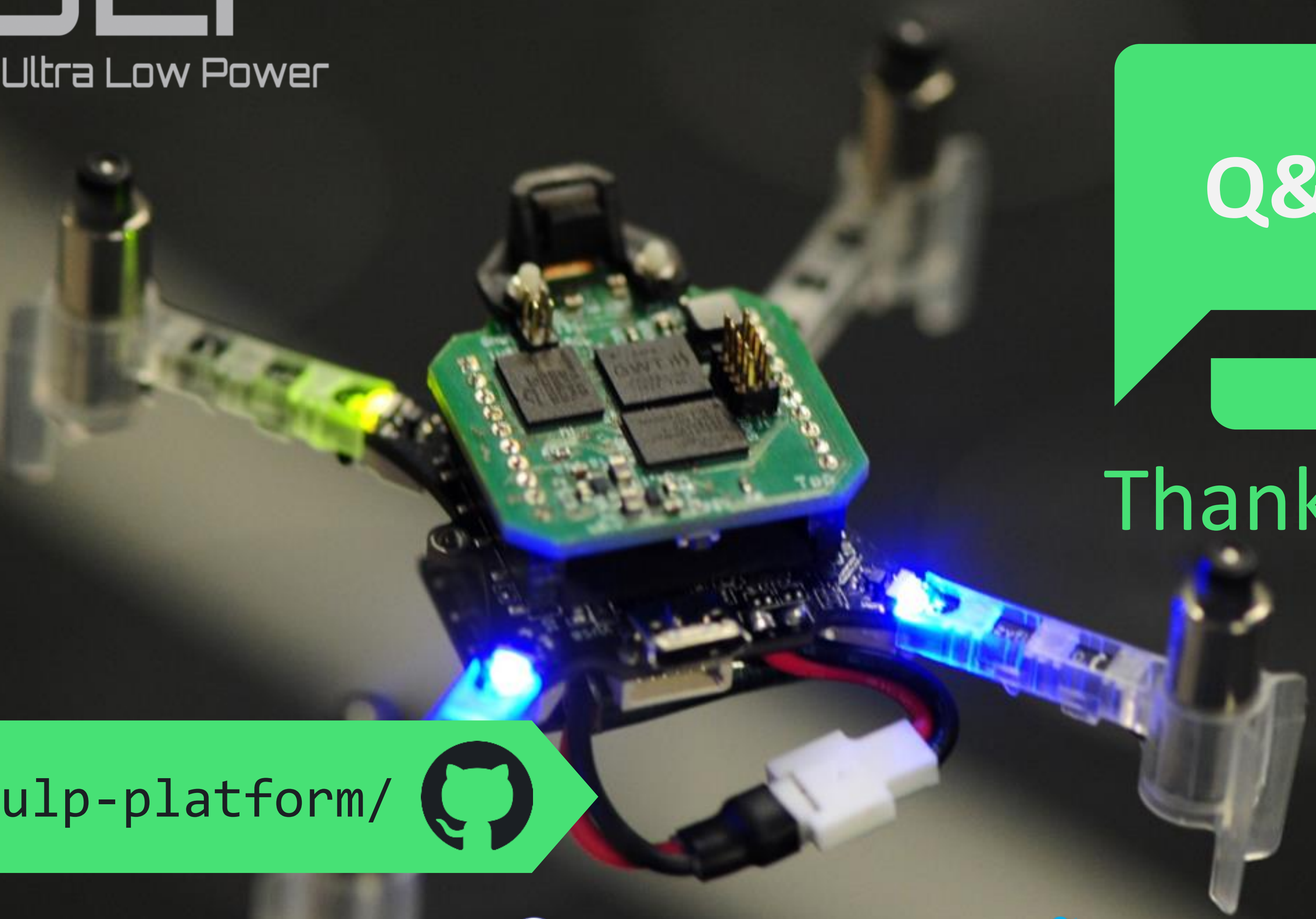

Q&A

Thank you!

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Image from Daniele Palossi